

```

name: default-deny
spec:
  selector: projectcalico.org/namespace != "kube-system"
  types:
    - Ingress
    - Egress

```

You can deploy the policy using the *calicoctl* command:

```
$ kubectl exec -ti -n kube-system calicoctl -- /calicoctl
apply -f - < deny.yml
```

Now, log into the busybox pod and try communicating with the nginx pod:

```
$ kubectl -n calico-demo exec busybox -it -- sh
$ wget -q -T 5 nginx -O -
```

The 'wget' command will fail with timeout since the nginx pod is not accessible.

Allowing egress traffic from the busybox pod: To allow egress traffic from the busybox pod, create a network policy as follows:

```

apiVersion: projectcalico.org/v3
kind: NetworkPolicy
metadata:
  name: busybox-egress
  namespace: calico-demo
spec:
  selector: run == 'busybox'
  types:
    - Egress
  egress:
    - action: Allow

```

You can deploy the policy using the *calicoctl* command:

```
$ kubectl exec -ti -n kube-system calicoctl -- /calicoctl
apply -f - < egress.busybox.yml
```

Test this by using the following command:

```
$ kubectl -n calico-demo exec busybox -it -- sh
$ wget -q -T 5 google.com -O -
```

This will give you a home page for Google.

But you will still not reach nginx since ingress traffic for nginx is blocked. For that we need to create an ingress traffic

network policy for the nginx pod.

Allowing ingress traffic for the nginx pod: To allow ingress traffic for nginx pod, we need to create a network policy as follows:

```

apiVersion: projectcalico.org/v3
kind: NetworkPolicy
metadata:
  name: nginx-ingress
  namespace: calico-demo
spec:
  selector: app == 'nginx'
  types:
    - Ingress
  ingress:
    - action: Allow
      source:
        selector: run == 'busybox'

```

This will allow ingress traffic to the nginx pod from the busybox pod. This can be activated with the following command:

```
$ kubectl exec -ti -n kube-system calicoctl -- /calicoctl
apply -f - < ingress.nginx.yml
```

You can verify this by using:

```
$ kubectl -n calico-demo exec busybox -it -- sh
$ wget -q -T 5 nginx -O -
```

The wget command will present us nginx's index page.

This demo shows how Calico can be used to manage Kubernetes networking. If you are interested in the latter, do explore other Calico capabilities from its official documentation. **END** 

References

- Official documentation
- <https://kubernetes.io/docs/concepts/cluster-administration/networking/>
- <https://projectcalico.docs.tigera.io/about/about-calico>
- <https://minikube.sigs.k8s.io/docs/>

By: Abhijeet Kasurde

The author works for Red Hat. He is a free and open source software evangelist who loves to explore new technologies and software.